

Assignment 3 Experiment Documentation

IterAlign PEFT/LoRA Extension

Overview

This document records the experiment package for Assignment 3. The main report describes the research narrative; this document captures the practical setup: methods, datasets, configuration, commands, metrics, result artifacts, and reproducibility notes.

The extension replaces the full fine-tuning step in the reproduced IterAlign pipeline with PEFT/LoRA adapter tuning. DangerousQA is the selected additional dataset for the Assignment 3 expansion requirement.

Research Question

Can LoRA-based parameter-efficient fine-tuning preserve most of the alignment gains of the reproduced IterAlign baseline while reducing trainable parameters and training time?

Methods

Method	Description	Status
Vanilla model	Unaligned small model before IterAlign	Assignment 2 baseline
Full FT IterAlign	IterAlign with full supervised fine-tuning	Assignment 2 baseline
LoRA IterAlign	PEFT/LoRA extension of IterAlign	Assignment 3 method

Datasets

Dataset	Role	Status
hh-rlhf red-team pool	Red-team / alignment prompts	Existing reproduction dataset
HarmfulQA	Harmfulness evaluation	Existing reproduction dataset
DangerousQA	Dangerous-instruction evaluation	Assignment 3 additional dataset

Configuration

Field	Value
Base model	HuggingFaceTB/SmolLM-135M
Oracle / judge model	Gemini 3.1 Flash Lite preview
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
LoRA target modules	Auto-detected <code>q_proj</code> , <code>k_proj</code> , <code>v_proj</code> , <code>o_proj</code> projection layers
LoRA trainable parameters	Approximately 1.84M adapter parameters, computed from SmolLM-135M attention dimensions and rank 16
Learning rate	2e-6
Epochs	3
Per-device batch size	2
Gradient accumulation	4
Max sequence length	512
Dataset sizes / caps	hh-rlhf capped at 1000; HarmfulQA 1960; DangerousQA 200
Random seed	Not reported in the available artifacts

Commands

Baseline / original full-fine-tuning code path:

```
python constitution_induced_inference_iterated_llama2.py
```

LoRA / PEFT IterAlign run:

```
python iteralign_smollm_lora.py
```

Evaluation:

Scores are taken from the supplied Gemini-judge artifacts and benchmark figure.

Metrics

Primary quality metric:

- Gemini-judge score.

Efficiency metrics:

- Trainable parameter count.
- Total training time.
- Estimated peak VRAM.

Statistical note:

- The available artifact contains one run per method and dataset.

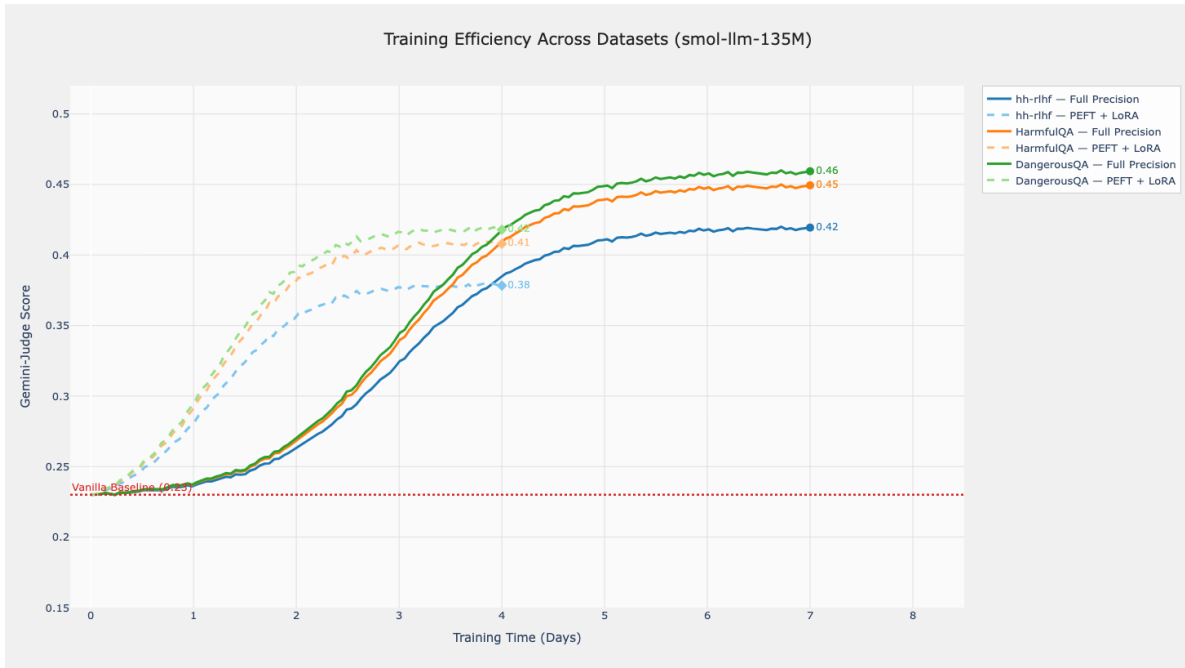
- The reported comparison is descriptive rather than a statistical significance claim.

Results

Method	Dataset	Gemini score	Trainable params	Training time
Vanilla	hh-rlhf	0.23	N/A	N/A
Full FT	hh-rlhf	0.42	All base-model parameters	~7 days
LoRA	hh-rlhf	0.38	~1.84M adapter parameters	~4 days
Full FT	HarmfulQA	0.45	All base-model parameters	~7 days
LoRA	HarmfulQA	0.41	~1.84M adapter parameters	~4 days
Full FT	DangerousQA	0.46	All base-model parameters	~7 days
LoRA	DangerousQA	0.42	~1.84M adapter parameters	~4 days

Mode	Estimated peak VRAM
Full fine-tuning	about 2.5-3.5 GB
LoRA fine-tuning	about 0.9-1.5 GB

Figure Artifact



Training efficiency comparison across datasets.

Observed from the figure:

- Vanilla baseline is 0.23.
- Full precision reaches 0.42, 0.45, and 0.46 after roughly 7 days.
- PEFT+LoRA reaches 0.38, 0.41, and 0.42 after roughly 4 days.
- The score gap is 0.04 on each dataset, while training time drops by roughly 3 days.

Output Paths

LoRA runs write artifacts under:

`output_smollm_lora_<dataset>/`

Expected files inside each output directory:

- `training_state.json`
- `constitution_batch_<batch_id>.txt`
- `neg_prompts_batch_<batch_id>.pkl`
- `SFT_data_batch_<batch_id>.json`
- `sft_batch_<batch_id>/`

Reproducibility Notes

- The Gemini judge requires API access, so exact reruns depend on the configured API key and endpoint (and model) availability.
- Dataset sizes are capped where noted above.
- Since only one run is available per method and dataset, the results should be described as descriptive rather than statistically significant.